

## 3. Requirements Engineering

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The development of a robust medical recommendation system necessitates a thorough understanding of the stakeholders involved and the technical constraints of the environment.

### 3.1 Stakeholder Analysis

- **Patient/User:** The primary end-user who interacts with the system to input symptoms. Their main need is an intuitive interface and accurate, easy-to-understand recommendations.
- **Administrator:** Responsible for maintaining the medical knowledge base. The administrator ensures that the datasets (medications, precautions) remain updated with current clinical standards.
- **Developer/Data Scientist:** The technical stakeholder focused on model training, optimizing algorithm performance, and ensuring the seamless integration of the backend ML logic with the frontend interface.

### 3.2 Functional Requirements

1. **Symptom Input:** The system must allow users to select multiple symptoms from a predefined list or through a search-based interface.
2. **Disease Prediction:** The core ML engine must process the input symptoms and output the most probable disease.
3. **Comprehensive Recommendation:** Upon prediction, the system must display:
  - The predicted disease name.
  - A detailed description of the disease.
  - Recommended medications.
  - Precautionary measures to be taken.
  - Suggested diet and workout plans [\[1\]](#).
4. **Authentication:** Secure login and registration for users to save their medical history.

### 3.3 Non-functional Requirements

- **Accuracy:** The ML model must maintain a high precision rate (ideally >90%) to ensure safety.
- **Usability:** The interface must be accessible to users with varying levels of technical literacy.
- **Latency:** Prediction results should be generated within a few seconds of symptom submission.
- **Reliability:** The system should provide a disclaimer that recommendations are for informational purposes only and not a substitute for professional diagnosis.